

Competent, Flexible, Safe Agents

Helpers We Can Trust

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Competent autonomous helper agents that inhabit the same worlds we do.

We need to trust them.

Behavior Alignment

Value alignment is behavior alignment.

- ▶ Narrow pursuit of task goals can be dangerous.
- ▶ Agents must respect our values.
- ▶ Value alignment manifests in behavior.

Notes:

- ▶ We do reinforcement learning (RL). The word “value” is used in many technical contexts in RL.
- ▶ Can be argued that “value alignment” is a misnomer.
 - Artificial agents only value things in a strict technical sense, not in the moral sense we mean when we talk about human values.

Design Goals for Safe Agents

Competence

- ▶ Safety mechanisms must preserve the agent's competence as much as possible.

Completeness

- ▶ Behavior alignment must take all values into account.

Flexibility

- ▶ We should be able to reuse values across task environments.

Explainability

- ▶ Trustworthy AI explains itself.

Technical Approaches

Learning

- ▶ We know from experience with expert systems the pitfalls of trying to enumerate all the necessary values.
- ▶ Core approach: learn our implicit values from explicit behavior.

Neurosymbolic AI

- ▶ Use symbolic representations where possible to support explainability.

The Bitter Lesson

- ▶ Must get the balance right.

How to learn alignment

Main idea: observe behavior, e.g., $\tau = (s_0, a_0, r_1, s_1, a_1, r_2, \dots, s_{t-1}, a_{t-1}, r_t)$

- ▶ Machine theory of mind (MToM)
 - ▶ Policy reconstruction
 - ▶ Plan recognition
- ▶ Offline learning
 - ▶ Permits scale
- ▶ Model-based reinforcement learning
 - ▶ Models permit incorporation of ideas from other fields like control theory, e.g., MPC

How to incorporate learned values

Agent architectures

- ▶ **Modular Reinforcement learning**
- ▶ Alignment critics

Algorithmic components

- ▶ **Action Restrictions**
- ▶ Causal models

Modular Reinforcement Learning

Single RL agent composed of modules that operate in parallel.

- ▶ Policies of each module $m_i \in M$ is $\pi_i : S \rightarrow A$
- ▶ Policy of the command arbitrator is $\pi_{CA} : S \rightarrow M$.
- ▶ Modules can use any kind of decision making algorithm
- ▶ Arbitrator, while initially designed using RL, can use any algorithm to choose modules.

Several appealing characteristics from a multiagent safety perspective.

- ▶ MRL agents are equally optimal to monolithic or Q-decomposition agents, i.e., MRL imposes no penalty on the agent's **competence**
- ▶ Arbitrator's policy tells us which module decides each action, affording **explainability**.
- ▶ Representation of an agent as composed of modules corresponds to idea that we constantly pursue both task goals and value goals simultaneously.

We are currently developing experiments and reports that demonstrate the **explainability** capabilities of MRL.

Generalizable Action Restriction Learning

Our first approach: **predicate-based action restriction learning (PARL)**

- ▶ Uses a factored representation of states containing boolean attributes.
- ▶ Learns from offline trajectories of teacher behavior.
- ▶ Employs MToM technique: policy reconstruction.
 - ▶ Identifies teacher's departures from a task-optimal policy.
- ▶ PARL algorithm learns restricted action types, which can be fed to any planning algorithm.
- ▶ An action type $\mathcal{T}$ is an ordered pair of active predicate sets characterizing the transition from state s_{t-1} to s : $\mathcal{T}_t = (\mathcal{P}_{t-1}, \mathcal{P}_t)$

Restricted action types expressed as predicates over two-step tuples of behavior trajectories, which indicate *kinds* of actions to avoid.

- ▶ PARL can learn action restrictions with a single teacher trajectory and generalize that learning to a different environment with compatible boolean state attributes, supporting **flexibility**.
- ▶ PARL combines learning with symbolic AI, supporting **explainability**.
- ▶ PARL is maximally optimal under action restrictions, resulting in **competent** agents.
- ▶ Because learning requires no explicit teaching other than collecting behavior trajectories, this approach supports **completeness** – as more behavior trajectories are collected, more values are represented.

A work-in-progress report with initial PARL results is currently under review for the Adaptive Learning Agents workshop at AAMAS 2026.